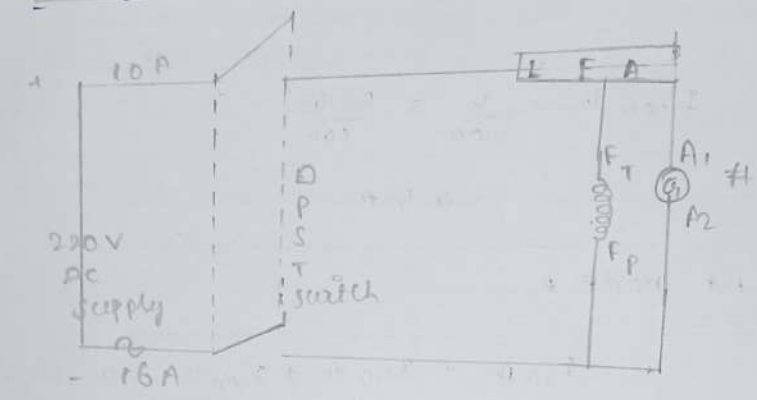
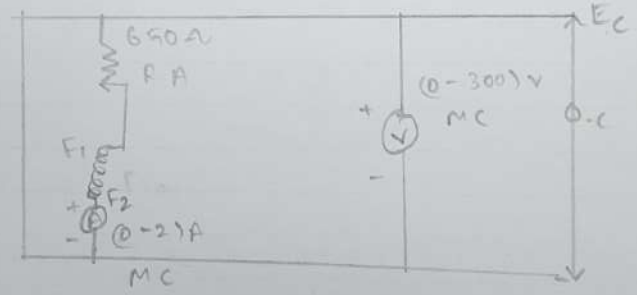
Open circuit characteristics:





OPEN CIRCUIT AND LOAD CHARACTERISTICS OF DC SHUNT GENERATOR.

Aim:

To draw the open circuit and load characteristics of DC shunt generator.

APPARATUS Required:

Laptop with internet connection.

Theory:

DC generator converts mechanical energy into electrical energy.

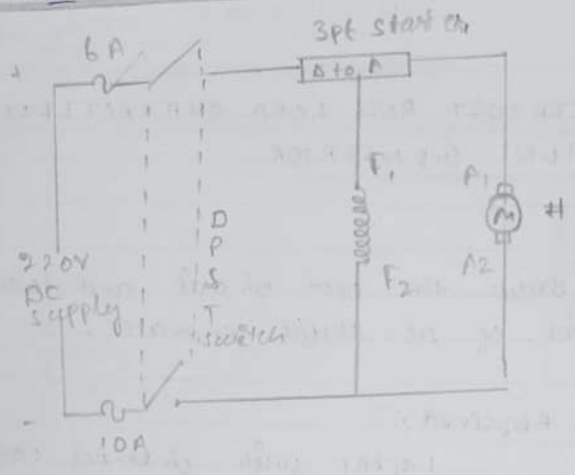
DC generator works on the principle of faradays law. It states that "when a conductor cuts the magnetic flux and emf is induced".

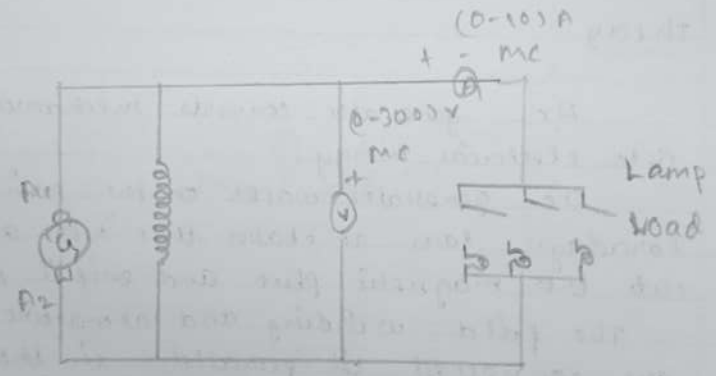
The field winding and armature winding are connected in parallel in the shunt generator.

Open circuit characteristics

It is also known as magnetisation characteristics or no load characteristics. It is graph drawn between open circuit voltage (E_0) and field current (I_F).

Load circuit characteristic:





Load circuit characteristics:

It is a graph drawn between load voltage (V_L) and load circuit (I_L). It is also known as external characteristics.

PROCEDURE:

Open circuit characteristics:

1. connections are made as per the circuit diagram.
2. Switch on the supply and note down the readings (voltage and current)
3. By varying the field rheost. at different values of open circuit voltage and field current are noted.
4. Plot the graph between open circuit voltage and field current.

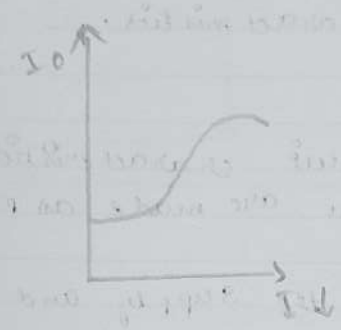
Load circuit characteristic:

1. connections are made as per the circuit voltage and current.
2. Switch on the supply and note the no load voltage and current.
3. By connecting different nodes, different values of load voltage and current.
4. Plot the graph between load voltage and load current.

Open circuit characteristics:

Model Graph:

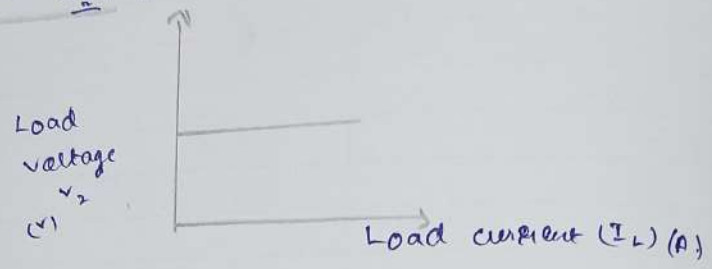
E_0 V, I_f



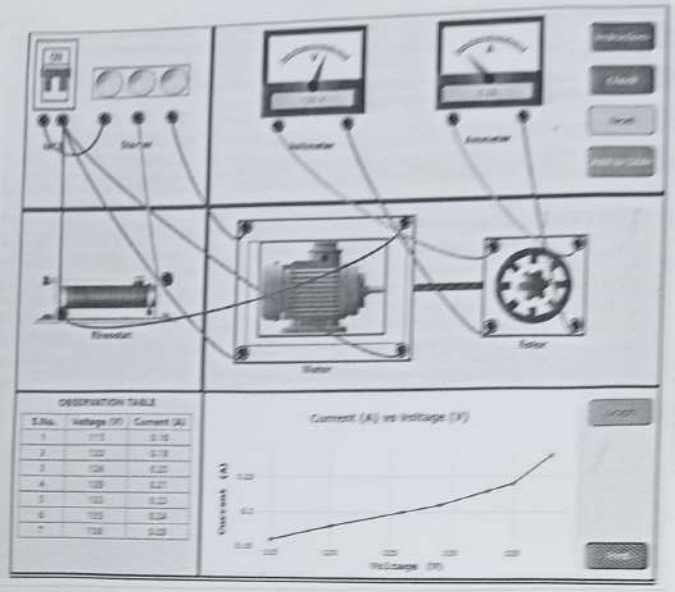
S.NO	Voltage(V)	Current(A)
1	115	0.16
2	120	0.18
3	126	0.20
4	129	0.22
5	133	0.23
6	135	0.24
7	138	0.25
8	142	0.30

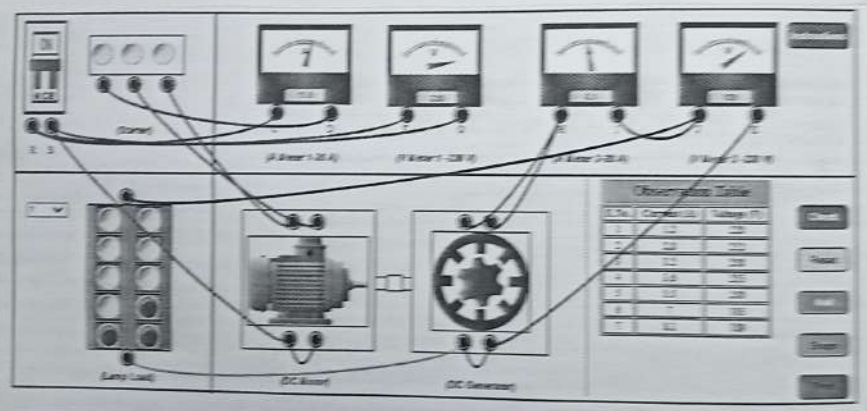
Load characteristics:

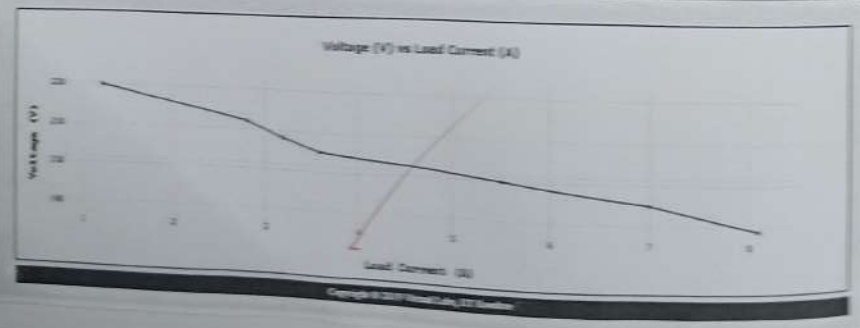
model Graph:



S.NO	Current (A)	Voltage (V)
1	1.2	2.20
2	2.8	2.12
3	3.2	2.08
4	3.6	2.05
5	5.5	200
6	7	195


OBSERVATION TABLE
S.No.
Voltage (V)
Current (A)
Current (A) vs Voltage (V)
Current (A)
Voltage (V)


(Lamp Load)
(DC Motor)
(DC Generator)


Voltage (V) vs Load Current (A)
Voltage (V)
Load Current (A)

100

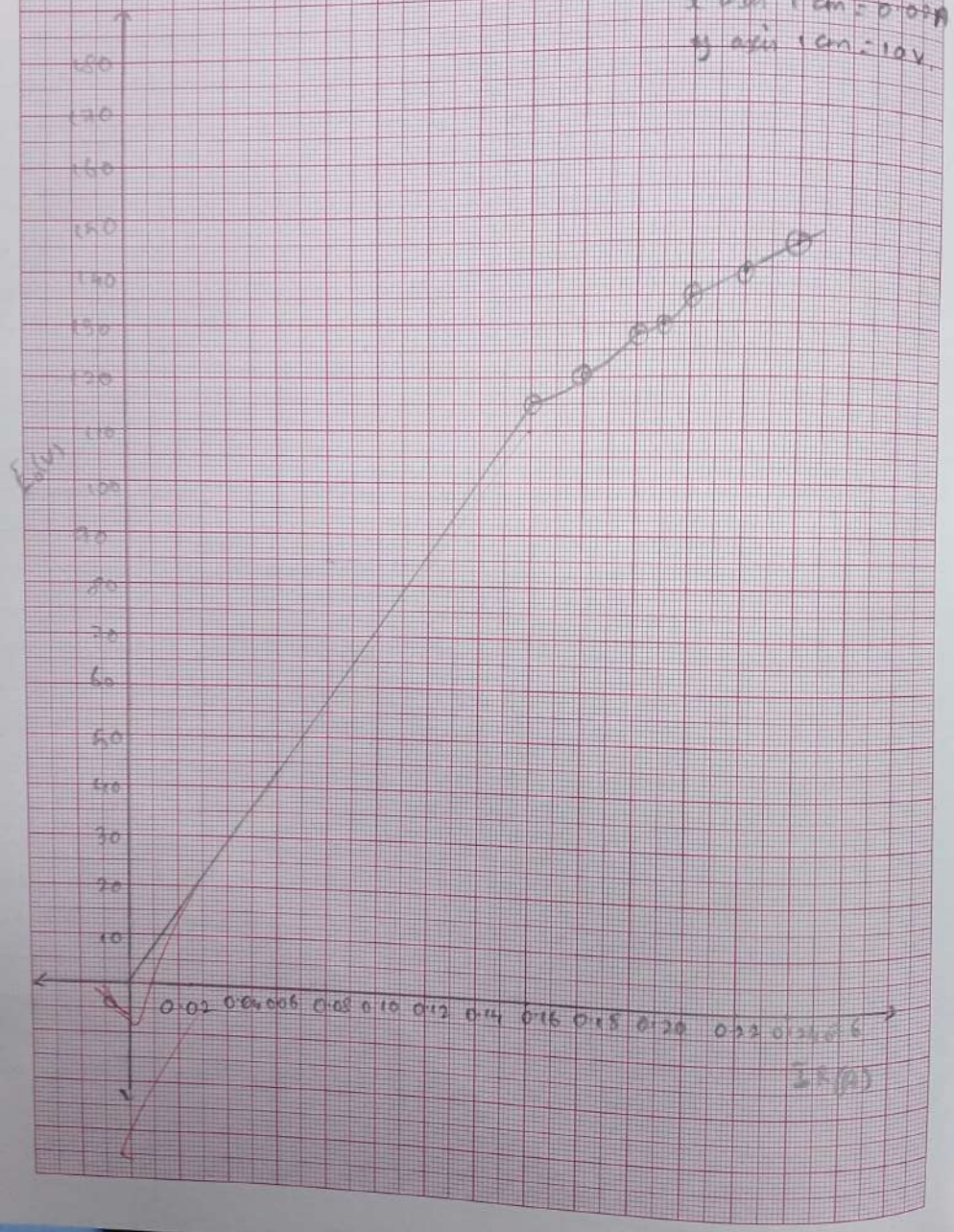
11/03/26

Result:

Thus, the open circuit and load characteristics of DC Shunt generator were verified successfully.

Open circuit characteristic

Scale:
x axis 1 cm = 0.01 A
y axis 1 cm = 10 V



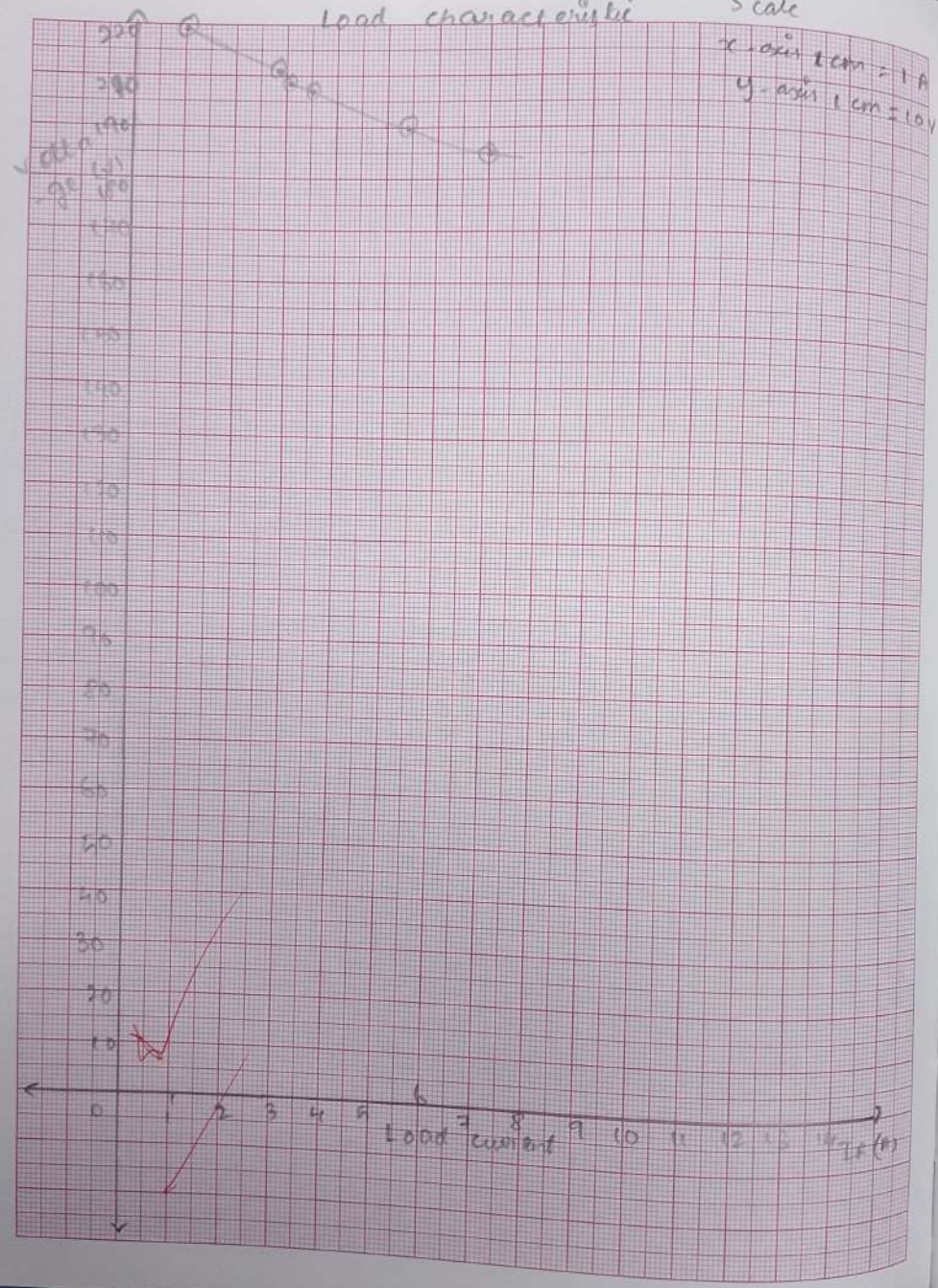
Expt. No.

Load characteristic

Scale

x-axis 1 cm = 1 A

y-axis 1 cm = 10 V



Expt. No.